

For data extraction on 7/12/2026

- **Ingestion run one**

Table	Records Inserted	Duplicates	Errors
App	5	0	0
Country	5	0	0
Raw_Reviews	10930	0	0
Cleaned_Reviews	8915	0	0

- **Ingestion run two**

Table	Records Inserted	Duplicates	Errors
App	0	5	0
Country	0	5	0
Raw_Reviews	0	10930	0
Cleaned_Reviews	0	8915	0

-The second ingestion run correctly identified all existing records and skipped duplicate insertions

- **Quality Flags (records needed to be removed)**

Sum(condition=True)

Duplicate Records	Missing Review Content	Non-English Review Content	Empty Review After Cleaning
0	0	2015	0

Missing Review Content

-Reviews with empty or placeholder values (e.g., "", "NA", "N/A", "Unknown") are flagged because they do not contain meaningful textual information.

Duplicate Records

-Duplicate reviews are identified using the "review_id". Only the first occurrence is retained, while subsequent duplicates are flagged to avoid repeated records in downstream analysis.

Non-English Review Content

-Since the project focuses on English-language text analysis, reviews whose detected language is not English are flagged for removal.

Empty Review After Cleaning

-After text preprocessing (such as removing URLs, emojis, punctuation, and other non-textual characters), some reviews may become empty. These records are also flagged because they no longer contain analyzable content.

The quality flags were designed to document why a raw review should be excluded from the **Cleaned_Reviews dataset** while preserving the original record for traceability.

Instead of deleting records immediately, each raw review is evaluated against a series of predefined quality checks. If a record fails a quality check, a removal reason is recorded in the 'record_remove_reason' field. This allows every removed record to remain traceable in the **Raw_Reviews table**.

- **Table relationship**

1. All foreign key relationships remained valid after the ingestion process. No orphan records were detected in any of the three relationships.

2. The database query results are consistent with the Python ETL Statistics:

Result Grid			Filter Rows:	
	app_name	COUNT(*)		
▶	Spotify	2205		
	YouTube	2205		
	Candy Crush	2205		
	Monopoly GO	2110		
	ChatGPT	2205		

app_name	
YouTube	2205
Candy Crush	2205
ChatGPT	2205
Spotify	2205
Monopoly GO	2110
Name: count, dtype: int64	

Result Grid			Filter Rows:	
	country_name	COUNT(*)		
▶	us	2205		
	gb	2205		
	au	2205		
	ca	2205		
	in	2110		

[43]: country_name	
us	2205
gb	2205
au	2205
ca	2205
in	2110
Name: count, dtype: int64	

	raw_without_cleaned
▶	2015

	review_cleaned
▶	8915

record_remove_reason	
None	8915
Non-English Review Content	2015
Name: count, dtype: int64	

2015 raw review records did not have corresponding records in Cleaned_Reviews because they were intentionally excluded during the cleaning stage due to non-English content.

For data extraction on 7/13/2026

- **Ingestion run one**

Table	Records Inserted	Duplicates	Errors
App	0	5	0
Country	0	5	0
Raw_Reviews	1	10931	0
Cleaned_Reviews	4	8919	0

- **Quality Flags**

Sum(condition=True)

Duplicate Records	Missing Review Content	Non-English Review Content	Empty Review After Cleaning
0	1	1997	0

```
record_remove_reason
None                8933
Non-English Review Content  1997
Missing Review Content      1
Name: count, dtype: int64
```

This time, the pipeline seemed to correctly skip duplicate records, as shown by the duplicate counts above. However, when new RSS data became available, I noticed that Raw_Reviews inserted one new record, while Cleaned_Reviews inserted four new records instead of one.

I checked the Python table records and the database records, and I found that for the same raw_review_pk, there are two different records in the Python pipeline and the database, respectively.

I think this happens because raw_review_pk is generated in Python based on the row order of the extracted dataset. However, the App Store RSS feed is dynamic, and the order of reviews can change between ingestion runs; the same row number may correspond to a different review in subsequent runs.

However, if raw_review_pk is generated automatically by SQL, I cannot populate the raw_review_pk foreign key in Cleaned_Reviews because those values are not available during the ETL process.

I would appreciate any design advice for addressing this issue.

Python:



```
]: raw_reviews_mysql[
    raw_reviews_mysql["review_id"] == "14274280739"
]
```

	raw_review_pk	review_id	app_id	country_id	author	rating	version	title	content	review_date	review_lang
8625	8626	14274280739	6448311069	5	s.wasim	5	1.2026.181	Na	Na	2026-07-07T07:43:32-07:00	

review_language	duplicate_flag	record_remove_reason
vi	False	Missing Review Content

Database:

Result Grid Filter Rows: Edit: Export/Imp					
	raw_review_pk	review_id	title	content	record_remove_reason
▶	8626	14272754263	Title		Non-English Review Content